PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Bridging the impactibility gap in population health management: a systematic review
AUTHORS	Orlowski, Andi; Snow, Sally; Humphreys, Heather; Smith, Wayne; Jones, Rebecca Siân; Ashton, Rachel; Buck, Jackie; Bottle, Alex

VERSION 1 – REVIEW

REVIEWER	Marafino, Ben Stanford University
REVIEW RETURNED	08-Jun-2021

GENERAL COMMENTS	Thank you for the opportunity to review this manuscript. It is clearly-written, timely, and addresses an important emerging topic—specifically, the use of impactibility modeling in population health management. Seeing this manuscript makes me wish that I had been able to read something like it when I first started out in this field. As such, I have no doubt that its publication will be of considerable interest to readers, particularly as impactibility-based methods begin to supplant risk-based ones in many PHM applications. Large health systems in the US—including Kaiser Permanente and others—are already starting to go down this road, and it will undoubtedly have a major impact on PHM, especially as value-based care models continue to proliferate. My major and minor comments are listed below. Major points 1. I feel it might be possible to expand the current scope of this review to more completely capture the extent of prior work on impactability modeling. In particular, consider including studies that aim to characterize heterogeneous treatment effects (HTEs) or other measures of benefit—e.g. Sussman et al (reference below) which assess risk-based HTE for an intervention intended to prevent progression to diabetes. Also consider including studies invoking "benefit models", "likelihood of benefit", "C statistic for benefit", etc. I realize this might be a bit tricky since the line between whether such studies qualify as PHM or not is somewhat blurred—not to mention if such studies are actually performing case finding, which appears to be an inclusion criterion (#7 of Table S2). However, HTE/benefit-based analyses of potentially population-scale interventions seem to be compatible with Lewis's description of an ideal impactibility model that "use[s] information on differential effects of a specific preventive intervention offered at random." Also, I understand that studies of specific diseases were excluded, which might disqualify this particular study—but this seems a bit arbitrary considering the inclusion of studies such as Steventon et al 2012

	which examined 3 chronic diseases, among others. Sussman JB, Kent DM, Nelson JP, Hayward RA. Improving diabetes prevention with benefit based tailored treatment: risk based reanalysis of Diabetes Prevention Program. BMJ. 2015;350:h454. Published 2015 Feb 19. doi:10.1136/bmj.h454 2. Related to the above, I think the working definition of impactibility needs to be made more concrete. Specifically, the case-finding component of impactibility seems a bit vague—I have usually viewed case-finding as an application of risk prediction models, i.e., that these models can be applied to identify potential cases of an adverse outcome over some time span and to prioritize a preventive intervention accordingly. But—and I could be wrong—I get the impression that the manuscript alludes to it as a step in the impactibility assessment process: (1) selecting potential responders from a wider, enrolled population at baseline, and then (2) prioritizing among these potential responders. This seems to be in conflict with the studies that were included since the populations analyzed do not appear to correspond to some "total" baseline population—be it all enrollees on a health plan or all patients on a GP surgery register for example. 3. Putting myself into the shoes of a reader who may be a newcomer to the impactibility field, it might be helpful to put together a more comprehensive methodological framework for these impactibility models. I found myself wondering what kind of methods were going into estimating these models—specifically, what were they attempting to estimate (if not outcome risk) and with what algorithms (just vanilla logistic regression or something more exotic)? This is particularly relevant for the PTS-based approaches since it is not clear how "success" is measured whereas the notion of outcome risk is easier to grasp. Altogether, this will depend on whether you decide to expand the scope of this review as well as on the precise definition of impactibility ultimately used in this review. 4. Figure 2: I found Figure 2 a bit confusing; I'm not sure if it communicates the intended message effectively. The one thing that Figure 2 does communicate well is that the impactible population is often much larger than just those individuals predicted to be at high risk. However, as the main text establishes at several points, high-risk individuals themselves exhibit varying degrees of impactibility and in some cases may be less impactible on average than those at lower risk. I think this concept could be better communicated here. Again, this will ultimately depend on how you define impactibility in this manuscript. In addition, I'm not sure how to interpret the axes in Figures 2A/B. I could be wrong, but I get the impression that the y-axis ("Experience of X over time Y") is intended to refer to outcome risk—that is, the risk of X. This could be clarified. Moreover, the x-axis ("Size of population") seems redundant as population size is already captured by the size of the strata. Furthermore, in figure 2B, "Impactibility A", "Impactibility B", and "Impactibility C" are not clearly defined. Are these meant to refer to different potential modes of impactibility? That is, perhaps an individual identified as "impactibility A" would be responsive to a corresponding intervention, and likewise for the B and C individuals. But do individuals A/B/C have, say, the same chronic condition, or

	do they have different problems altogether? I think this also relates to my point about the “case finding” bit in #1 and #2 and whether it is a necessary component of the working definition of impactibility used here—if these individuals are meant to be “fished out” of the population at baseline or are assessed for impactibility based on some qualifying event; for example, a hospital discharge. Going back to communicating the key ideas, one suggestion—given a hypothetical preventative intervention—is to mock up a plot of treatment effect as a function of predicted outcome risk. This usually ends up looking like something like a J- or U-shaped curve where the treatment effect is less for individuals at higher risk compared to those in some lower-risk strata. You can see an example in Figure 1 of our paper referenced below—although you would not have to show the full treatment effect distribution within each risk stratum like we did; just mock up a smooth function of predicted outcome risk. In addition, you can present different shapes of this function to show different scenarios, such as where the treatment effect correlates with predicted risk ("risk magnification") and in which case prioritizing based on predicted risk would in fact be effective. This can be presented alongside a mocked histogram of the predicted risk distribution (which very often is right-skewed) to drive the point home. I find this construct quite useful and I have used it often in talks where it tends to resonate with audiences. Marafino BJ, Schuler A, Liu VX, Escobar GJ, Baiocchi M. Predicting preventable hospital readmissions with causal machine learning. Health Serv Res. 2020;55(6):993-1002. doi:10.1111/1475-6773.13586 5. Figure 3: The second box (“Bridge the impactibility gap”) is a bit vague. I think this could be broken down (and the other boxes could be revised as well) to explain what goes into bridging the impactibility gap and would help with the framing of the definition of impactibility used in the paper. 6. Consider adding to Table 1 as limitations of PTS and behavioural response models (as well as HTE/benefit-based approaches if you decide to go down that road) the need for interventional data—as opposed to retrospective data for a risk model. These interventional data can be observational or randomized; the former type of data come with an additional limitation in that they need to be deconfounded. Minor points The first sentence reads a bit awkwardly (“in which”)—I think it could be revised to get the paper off to a stronger start. Would help to concisely define impactibility just before the first mention on page 4, line 46. page 2, line 27: “For health conditions”, should be “For the health conditions theme”? page 2, lines 38-39 - I couldn't parse this: “In the clinical judgment theme suggested a complementary role for models” page 2, line 46 - “high models scores” should be “high model

	scores"? Page 3, line 10 - fragment? Was this meant to conclude the sentence immediately before it? "Limitation—as this is a growing area of interest…" page 5, line 50 - "a high risk" page 6, lines 20-22 - "as this as", "ethnics approval" page 8, line 40 - "combined approached". Also, perhaps briefly mention why the average risk score increased. Were lower-risk high-risk patients filtered out as a result of applying clinical judgement? page 8, line 44, "muiltimorbid" page 10, line 7 - missing a comma after "sociodemographic factors"? page 13, line 3 - "The areas most studied so far is PTS"; should be "area" Table 1 - under "Health needs/gap analysis", is "Suitable data to assess gaps are rarely available in real-world records" meant to be a limitation and not an advantage?

REVIEWER	Buja, Alessandra University of Padua
REVIEW RETURNED	10-Jun-2021

GENERAL COMMENTS	This systematic review aimed to feel a knowledge gap analysing studies, using different methodologies (and terminology), applied to evaluate impactability models in population health management. This issue could be very relevant for policy makers however, this manuscript, to be more useful for readers and potential users, should be revised mainly in the discussion section. Some part of the discussion is a succession of scattered and generic concepts more entailed for an introduction section. The discussion will benefit of a more structured framework in which each group of studies, by type of impactability models addressed, should be discussed to explain how previous studies' findings can be used in policy-making and in policy formulation in terms of health, health equity and social and economic gain. Moreover for each group of studies the strategies to improve health care researches about this issue could be discussed in the light of the main limits of the studied included (for example in the methods you revealed a problem with a non consistent terminology among studies). Finally, you can report also the remaining knowledge gaps to drive further new researches in this field.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Ben Marafino, Stanford University
Comments to the Author:
Thank you for the opportunity to review this manuscript. It is clearly-written, timely, and addresses an important emerging topic—specifically, the use of impactibility modeling in population health management. Seeing this manuscript makes me wish that I had been able to read something like it

when I first started out in this field. As such, I have no doubt that its publication will be of considerable interest to readers, particularly as impactibility-based methods begin to supplant risk-based ones in many PHM applications. Large health systems in the US—including Kaiser Permanente and others—are already starting to go down this road, and it will undoubtedly have a major impact on PHM, especially as value-based care models continue to proliferate. My major and minor comments are listed below.

- *We thank this reviewer for his thoughtful review.*

Major points

1. I feel it might be possible to expand the current scope of this review to more completely capture the extent of prior work on impactability modeling. In particular, consider including studies that aim to characterize heterogeneous treatment effects (HTEs) or other measures of benefit—e.g. Sussman et al (reference below) which assess risk-based HTE for an intervention intended to prevent progression to diabetes. Also consider including studies invoking "benefit models", "likelihood of benefit", "C statistic for benefit", etc.

I realize this might be a bit tricky since the line between whether such studies qualify as PHM or not is somewhat blurred—not to mention if such studies are actually performing case finding, which appears to be an inclusion criterion (#7 of Table S2). However, HTE/benefit-based analyses of potentially population-scale interventions seem to be compatible with Lewis's description of an ideal impactibility model that "use[s] information on differential effects of a specific preventive intervention offered at random." Also, I understand that studies of specific diseases were excluded, which might disqualify this particular study—but this seems a bit arbitrary considering the inclusion of studies such as Steventon et al 2012 which examined 3 chronic diseases, among others.

Sussman JB, Kent DM, Nelson JP, Hayward RA. Improving diabetes prevention with benefit based tailored treatment: risk based reanalysis of Diabetes Prevention Program. BMJ. 2015;350:h454. Published 2015 Feb 19. doi:10.1136/bmj.h454

- *We thank the reviewer for consideration the scope of our review. The aim of this literature review was to investigate whether impactibility modelling is being used and, if so, how rather than to test success or failure in relation to specific disorders/situations. Further work we have undertaken indicates that considering specific diseases, outcomes, and effects would be far beyond the scope of this review. We would therefore prefer to address these in further papers where each can be given sufficient space for exploration and discussion.*

  We felt it was appropriate to include the Steventon papers and others in the "Health conditions amenable to preventive care" category because the concepts they consider are how interventions targeted after the diagnosis of conditions (ie, any diagnosed condition, although the studies used certain ones as examples) have effects at the population level (mortality, hospital admissions, costs).

  We have clarified the high-level aim of this review at the end of the introduction. "The aim of this current study was to describe broadly how and in what contexts impactibility modelling has been implemented or assessed in PHM since 2010. It was beyond the scope of this review to consider how the approach has affected management of individual diseases, heterogeneity in treatment effects, and different types of health system."

  We have also altered the wording in the Results on p 6 to "Several studies inferred participants' potential to benefit from preventive care if it is targeted after they have received a diagnosis of a specific health condition[18–21] or if they have a multi-morbid cluster of health conditions.[22,23]"

2. Related to the above, I think the working definition of impactibility needs to be made more concrete. Specifically, the case-finding component of impactibility seems a bit vague—I have usually viewed case-finding as an application of risk prediction models, i.e., that these models can be applied to identify potential cases of an adverse outcome over some time span and to prioritize a preventive

intervention accordingly. But—and I could be wrong—I get the impression that the manuscript alludes to it as a step in the impactibility assessment process: (1) selecting potential responders from a wider, enrolled population at baseline, and then (2) prioritizing among these potential responders. This seems to be in conflict with the studies that were included since the populations analyzed do not appear to correspond to some "total" baseline population—be it all enrollees on a health plan or all patients on a GP surgery register for example.

- *Thank you for raising this point about case finding. Case finding is intended to be performed after the impactibility analysis indicates which patients will be most amenable to the treatment strategy being offered. We have adjusted figure 3 to make this clearer.*

3. Putting myself into the shoes of a reader who may be a newcomer to the impactibility field, it might be helpful to put together a more comprehensive methodological framework for these impactibility models. I found myself wondering what kind of methods were going into estimating these models—specifically, what were they attempting to estimate (if not outcome risk) and with what algorithms (just vanilla logistic regression or something more exotic)? This is particularly relevant for the PTS-based approaches since it is not clear how "success" is measured whereas the notion of outcome risk is easier to grasp. Altogether, this will depend on whether you decide to expand the scope of this review as well as on the precise definition of impactibility ultimately used in this review.

- *Thank you for this point. Most of the studies used logistic regression to identify impactible factors that would maximise engagement and quality of care and reduce costs aimed at "success" was indicated by criteria were not clearly established, as these would be set by the pepo. However, to make the aims and methods clearer to readers, we have added relevant details to Appendix Table S4.*

4. Figure 2: I found Figure 2 a bit confusing; I'm not sure if it communicates the intended message effectively. The one thing that Figure 2 does communicate well is that the impactible population is often much larger than just those individuals predicted to be at high risk. However, as the main text establishes at several points, high-risk individuals themselves exhibit varying degrees of impactibility and in some cases may be less impactible on average than those at lower risk. I think this concept could be better communicated here. Again, this will ultimately depend on how you define impactibility in this manuscript.

In addition, I'm not sure how to interpret the axes in Figures 2A/B. I could be wrong, but I get the impression that the y-axis ("Experience of X over time Y") is intended to refer to outcome risk—that is, the risk of X. This could be clarified. Moreover, the x-axis ("Size of population") seems redundant as population size is already captured by the size of the strata.

Furthermore, in figure 2B, "Impactibility A", "Impactibility B", and "Impactibility C" are not clearly defined. Are these meant to refer to different potential modes of impactibility? That is, perhaps an individual identified as "impactibility A" would be responsive to a corresponding intervention, and likewise for the B and C individuals. But do individuals A/B/C have, say, the same chronic condition, or do they have different problems altogether? I think this also relates to my point about the "case finding" bit in #1 and #2 and whether it is a necessary component of the working definition of impactibility used here—if these individuals are meant to be "fished out" of the population at baseline or are assessed for impactibility based on some qualifying event; for example, a hospital discharge.

Going back to communicating the key ideas, one suggestion—given a hypothetical preventative intervention—is to mock up a plot of treatment effect as a function of predicted outcome risk. This usually ends up looking like something like a J- or U-shaped curve where the treatment effect is less for individuals at higher risk compared to those in some lower-risk strata. You can see an example in Figure 1 of our paper referenced below—although you would not have to show the full treatment effect distribution within each risk stratum like we did; just mock up a smooth function of predicted outcome risk. In addition, you can present different shapes of this function to show different scenarios, such as where the treatment effect correlates with predicted risk ("risk magnification") and in which case prioritizing based on predicted risk would in fact be effective. This can be presented alongside a mocked histogram of the predicted risk distribution (which very often is right-skewed) to drive the point home. I find this construct quite useful and I have used it often in talks where it tends to resonate with audiences.

Marafino BJ, Schuler A, Liu VX, Escobar GJ, Baiocchi M. Predicting preventable hospital readmissions with causal machine learning. Health Serv Res. 2020;55(6):993-1002. doi:10.1111/1475-6773.13586

- *We thank the reviewer for these helpful comments and have updated the labelling in figures 2A and 2B. We have also reworded the figure legend and text around the figure citation in the main text and redistributed the dots in the figure to clarify that amenability is not determined by risk. We have not created new charts showing treatment effect because we are not exploring outcomes in this review. Rather, we are looking at impactibility analysis from a higher level (please see our response to point 2).*

5. Figure 3: The second box ("Bridge the impactibility gap") is a bit vague. I think this could be broken down (and the other boxes could be revised as well) to explain what goes into bridging the impactibility gap and would help with the framing of the definition of impactibility used in the paper.

- *Please see our response to point 2, along with additional adjustments made in response to this comment. We hope that figure 3 is now clearer and more informative.*

6. Consider adding to Table 1 as limitations of PTS and behavioural response models (as well as HTE/benefit-based approaches if you decide to go down that road) the need for interventional data—as opposed to retrospective data for a risk model. These interventional data can be observational or randomized; the former type of data come with an additional limitation in that they need to be deconfounded.

- *Thank you for this point. We have added a note "Require interventional data rather than retrospective patient data" to table 1*

Minor points

The first sentence reads a bit awkwardly ("in which")—I think it could be revised to get the paper off to a stronger start.

- *We have reworded the sentence.*

Would help to concisely define impactibility just before the first mention on page 4, line 46.

- *We have added Lewis's definition of impactibility models and included a reference.*

page 2, line 27: "For health conditions", should be "For the health conditions theme"?

- *Thank you for pointing out this error. We have reworded the sentence.*

page 2, lines 38-39 - I couldn't parse this: "In the clinical judgment theme suggested a complementary role for models"

- *Thank you for pointing out this error. We have reworded the sentence.*

page 2, line 46 - "high models scores" should be "high model scores"?

- *Thank you for pointing out this error. We have reworded the sentence.*

Page 3, line 10 - fragment? Was this meant to conclude the sentence immediately before it? "Limitation—as this is a growing area of interest…"

- *Thank you for pointing out this error. We have reworded the sentence.*

page 5, line 50 - "a high risk"

- *Thank you for pointing out this error. We have reworded the sentence.*

page 6, lines 20-22 - “as this as”, “ethnics approval”

- *Thank you for pointing out this error. We have reworded the sentence.*

page 8, line 40 - “combined approached”. Also, perhaps briefly mention why the average risk score increased. Were lower-risk high-risk patients filtered out as a result of applying clinical judgement?

- *Yes, the combination of risk score and clinical judgment helped to improve selection. We have added some wording to explain this more clearly.*

page 8, line 44, “muiltimorbid”

- *Thank you for pointing out this error. We have corrected the spelling.*

page 10, line 7 - missing a comma after “sociodemographic factors”?

- *Thank you for pointing out this error. We have added commas.*

page 13, line 3 - “The areas most studied so far is PTS”; should be “area”

- *Thank you for pointing out this error. We have reworded the sentence.*

Table 1 - under “Health needs/gap analysis”, is “Suitable data to assess gaps are rarely available in real-world records” meant to be a limitation and not an advantage?

- *Thank you for pointing out this error. We have moved the wording into the limitations column.*

Reviewer: 2
Dr. Alessandra Buja, University of Padua Comments to the Author:
This systematic review aimed to feel a knowledge gap analysing studies, using different methodologies (and terminology), applied to evaluate impactability models in population health management. This issue could be very relevant for policy makers however, this manuscript, to be more useful for readers and potential users, should be revised mainly in the discussion section.

Some part of the discussion is a succession of scattered and generic concepts more entailed for an introduction section. The discussion will benefit of a more structured framework in which each group of studies, by type of impactability models addressed, should be discussed to explain how previous studies’ findings can be used in policy-making and in policy formulation in terms of health, health equity and social and economic gain. Moreover for each group of studies the strategies to improve health care researches about this issue could be discussed in the light of the main limits of the studied included (for example in the methods you revealed a problem with a non consistent terminology among studies). Finally, you can report also the remaining knowledge gaps to drive further new researches in this field.

- *We thank this reviewer for her thoughtful comments. We have reworked the discussion in particular in an attempt to improve the manuscript in line with the suggestions made.*

VERSION 2 – REVIEW

REVIEWER	Marafino, Ben Stanford University
REVIEW RETURNED	06-Oct-2021

GENERAL COMMENTS	Thank you again for allowing me the opportunity to review the

	revised version of this manuscript. It appears that the authors have attended well to our comments and suggestions. In particular, the "risk stratification versus impactibility" discussion is well done. Some nitpicks, mostly having to do with style, but which I think the authors might find helpful (and which may be recommended by the technical editor anyway)— 1. In the abstract, I would try to set apart the themes from the rest of the text. It took me at least two reads to parse the text, which is not ideal. Perhaps use italics or "scare quotes", so "health conditions amenable to care", etc. Also, perhaps "For health conditions the theme" should be "For the theme health conditions..."? 2. "A model used to identify patients appropriate for a proactive programme of multimorbid care management showed reasonable concordance with physicians..." - not sure why you singled out these two models in the abstract. I know there are space limitations but perhaps highlight what made these two particularly notable. 3. page 4, line 46 - “time” should be “type” 4. page 5, line 24 - “how impactibility modelling might affect … different types of health system”. Not sure what is meant by this. Perhaps you meant to allude to this kind of modelling having possibly differential effects in different types of systems? 5. page 6, line 33 - remove the “as” after is 6. page 12, lines 12-13 - The 3 categories listed here sound a bit discordant with those listed in Table 1. Specifically the category “models concerned with HCP input” is listed as “comparison or combination with clinical judgement” in Table 1. Maybe use “integrating” instead of “concerned”? Would probably be fine either way—it’s just something to consider for the sake of consistency. 7. page 15, line 21 - “everyone” sounds a bit casual here. Perhaps “all people” (in keeping with the paper’s style to use “people” as opposed to “patients”) instead?

VERSION 2 – AUTHOR RESPONSE

Reviewer: 1
Dr. Ben Marafino, Stanford University
Comments to the Author:
Thank you again for allowing me the opportunity to review the revised version of this manuscript. It appears that the authors have attended well to our comments and suggestions. In particular, the "risk stratification versus impactibility" discussion is well done.

Some nitpicks, mostly having to do with style, but which I think the authors might find helpful (and which may be recommended by the technical editor anyway)—

- *As previously, we thank this reviewer for the time and effort expended and for the helpful comments.*

1. In the abstract, I would try to set apart the themes from the rest of the text. It took me at least two reads to parse the text, which is not ideal. Perhaps use italics or "scare quotes", so "health conditions amenable to care", etc. Also, perhaps "For health conditions the theme" should be "For the theme health conditions..."?

- *We have added double quotes around the theme categories and made appropriate changes to wording to aid readers.*

2. "A model used to identify patients appropriate for a proactive programme of multimorbid care management showed reasonable concordance with physicians..." - not sure why you singled out these two models in the abstract. I know there are space limitations but perhaps highlight what made these two particularly notable.

- *These two studies had the most concrete results to report in the abstract, but we agree that it was unclear why they were selected. We have reworded this part of the abstract to address the role of models in relation to clinical decisions more clearly.*

3. page 4, line 46 - “time” should be “type”

- *Thank you for spotting this error. We have made this change.*

4. page 5, line 24 - “how impactibility modelling might affect … different types of health system”. Not sure what is meant by this. Perhaps you meant to allude to this kind of modelling having possibly differential effects in different types of systems?

- *Thank you for pointing out this ambiguity. We have changed “systems” to “programmes” and hope that this is clearer.*

5. page 6, line 33 - remove the “as” after is

- *Thank you for spotting this error. We have made this change.*

6. page 12, lines 12-13 - The 3 categories listed here sound a bit discordant with those listed in Table 1. Specifically the category “models concerned with HCP input” is listed as “comparison or combination with clinical judgement” in Table 1. Maybe use “integrating” instead of “concerned”? Would probably be fine either way—it’s just something to consider for the sake of consistency.

- *Thank you for spotting this error. We have altered the wording to improve consistency.*

7. page 15, line 21 - “everyone” sounds a bit casual here. Perhaps “all people” (in keeping with the paper’s style to use “people” as opposed to “patients”) instead?

- *Thank you for this observation. We have changed the wording to “all people”.*